

ECG Heartbeat Categorization

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I. INTRODUCTION

Electrocardiogram (ECG) signals are one of the most commonly used biomedical signals for monitoring cardiac activity and assessing the condition of the cardiovascular system. By recording the electrical behavior of the heart, ECG signals allow clinicians to identify abnormal heart rhythms and various cardiac disorders such as arrhythmias. However, the continuous acquisition of ECG data in both clinical and ambulatory settings produces a large volume of signals, making manual analysis labor-intensive and prone to subjective errors.

With the rapid development of machine learning and deep learning techniques, automatic ECG signal analysis has gained increasing attention. These methods are capable of learning discriminative patterns directly from data and can assist medical professionals by providing reliable and consistent heartbeat classification.

In this practical work, ECG heartbeat categorization is studied using a publicly available dataset from Kaggle. Two different approaches are implemented and compared: a traditional machine learning model based on Random Forest, and a deep learning model using a one-dimensional Convolutional Neural Network (1D-CNN). Special attention is paid to the problem of class imbalance, which is a common issue in medical datasets. The performance of the proposed methods is evaluated and compared with the results reported in the original reference paper.

II. DATASET OVERVIEW

The dataset used in this study is the ECG Heartbeat Categorization Dataset available on Kaggle. It is derived from the MIT-BIH Arrhythmia Database and contains preprocessed ECG heartbeats extracted from raw recordings.

The dataset is divided into two subsets:

- **Training set:** 87,554 samples.
- **Testing set:** 21,892 samples.

Each sample corresponds to a single ECG heartbeat represented as a one-dimensional time-series vector of length 187. The last column of each sample indicates the class label. The heartbeats are categorized into five classes according to the AAMI standard:

- 1) Normal
- 2) Supraventricular
- 3) Ventricular
- 4) Fusion
- 5) Unknown

All ECG signals are normalized and padded to a fixed length, allowing them to be directly used as inputs for machine learning and deep learning models.

III. CLASS DISTRIBUTION

An important characteristic of the training dataset is the significant class imbalance. The number of samples in each class is summarized as follows:

- Normal: 72,471 samples
- Supraventricular: 2,223 samples
- Ventricular: 5,788 samples
- Fusion: 641 samples
- Unknown: 6,431 samples

As illustrated in Fig. 1, the Normal class dominates the dataset, while Supraventricular and Fusion beats are relatively rare. This imbalance can negatively affect classification performance, especially for minority classes, and therefore must be considered during model training and evaluation.



Fig. 1. Class distribution of the training dataset, highlighting the imbalance between different heartbeat categories.

IV. METHODOLOGY

A. Approach 1: Random Forest with GridSearchCV

As a baseline machine learning method, a Random Forest classifier is implemented. Random Forest is an ensemble-based algorithm that combines multiple decision trees and is known for its robustness and good performance on structured data.

To optimize the model, GridSearchCV is employed to search for the best hyperparameters. In order to reduce the impact of class imbalance during training, the weighted F1-score (`f1_weighted`) is used as the optimization metric. The optimal parameters obtained through this process are:

- Number of trees: 100
- Maximum depth: None
- Minimum samples split: 5

B. Approach 2: 1D-CNN with SMOTE

To further improve classification performance, especially for minority classes, a deep learning approach based on a

one-dimensional Convolutional Neural Network (1D-CNN) is adopted. Since ECG signals are temporal sequences, 1D convolutions are well-suited for extracting local temporal features from heartbeats.

Before training the CNN model, the Synthetic Minority Over-sampling Technique (SMOTE) is applied to the training data in order to balance the class distribution. By generating synthetic samples for underrepresented classes, SMOTE helps the model learn more representative decision boundaries.

The CNN architecture consists of multiple convolutional and pooling layers, followed by fully connected layers and a softmax output layer for five-class classification.

V. RESULTS

A. Random Forest Performance

The Random Forest model achieves a test accuracy of **97.42%**. Although the overall accuracy is high, the classification report reveals that the model struggles to correctly identify minority classes. In particular, the recall for Supraventricular and Fusion beats is relatively low, which can be attributed to the strong class imbalance in the dataset.

TABLE I
RANDOM FOREST CLASSIFICATION REPORT

Class	Precision	Recall	F1-score	Support
Normal	0.97	1.00	0.99	18118
Supraventricular	0.95	0.60	0.74	556
Ventricular	0.98	0.89	0.93	1448
Fusion	0.82	0.61	0.70	162
Unknown	1.00	0.95	0.97	1608
Accuracy			0.97	21892
Weighted Avg	0.97	0.97	0.97	21892

The confusion matrix for the Random Forest model is shown in Fig. 2, illustrating frequent misclassification of rare heartbeat categories.

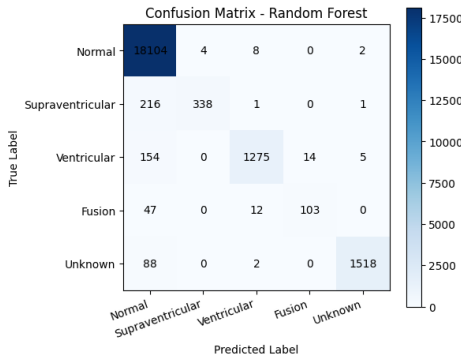


Fig. 2. Confusion matrix of the Random Forest model.

B. CNN Performance with SMOTE

After applying SMOTE and training the 1D-CNN model, the test accuracy increases to **98.36%**. More importantly, the recall for minority classes improves significantly. For example, the recall of Supraventricular beats increases from approximately 0.60 to 0.82, and the recall of Fusion beats increases from approximately 0.61 to 0.83.

TABLE II
RANDOM FOREST CLASSIFICATION REPORT

Class	Precision	Recall	F1-score	Support
Normal	0.97	1.00	0.99	18118
Supraventricular	0.95	0.60	0.74	556
Ventricular	0.98	0.89	0.93	1448
Fusion	0.82	0.61	0.70	162
Unknown	1.00	0.95	0.97	1608
Accuracy			0.97	21892
Weighted Avg	0.97	0.97	0.97	21892

The confusion matrix of the CNN model, presented in Fig. 3, confirms that the CNN combined with SMOTE provides better discrimination across all heartbeat categories.

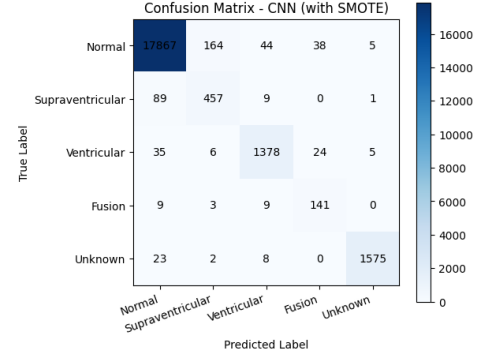


Fig. 3. Confusion matrix of the CNN model trained with SMOTE.

VI. CONCLUSION

In this practical work, ECG heartbeat categorization is addressed using both traditional machine learning and deep learning approaches. While the Random Forest model achieves strong overall accuracy, it shows clear limitations in detecting rare heartbeat classes due to data imbalance. By applying SMOTE and using a 1D-CNN architecture, both the overall accuracy and the recall of minority classes are significantly improved.

These results highlight the importance of handling class imbalance when designing machine learning systems for medical applications and demonstrate the effectiveness of deep learning methods for ECG signal classification.